UNIVERSITY END SEMESTER EXAMINATION

B.E. Computer Engineering - Semester VII

CE-701 Blockchain Technology Max Marks: 80 Time: 3 Hours

Q1. (a) Explain the structure of a block and how blocks are linked using hashes. [10]

(b) Compare Proof of Work and Proof of Stake consensus mechanisms. [10]

Q2. (a) What is a Merkle tree? Explain its role in SPV. [10]

(b) Write a Solidity smart contract for a simple voting system. [10]

Q3. (a) Explain the Ethereum Virtual Machine and gas. [10]

(b) Discuss the double-spending problem and how Bitcoin solves it. [10]

Q4. (a) Explain Hyperledger Fabric architecture. [10]

(b) Describe any two real-world applications of blockchain. [10]